

Quantum knot invariants and fiberedness

Lara San Martín Suárez (Caltech)

based on joint work with P. Orland, T. Saunders-A'Court, J. Svoboda

Caltech/USC Algebra & Geometry Seminar

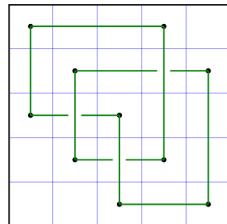
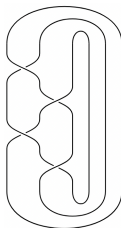
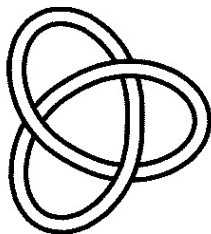
November 20, 2025

Motivation

Quantum invariants of knots

Let $K \subset S^3$ be a knot (embedding of a circle in the 3-sphere). Represent K in the plane by a knot diagram D .

Commonly used knot diagrams:

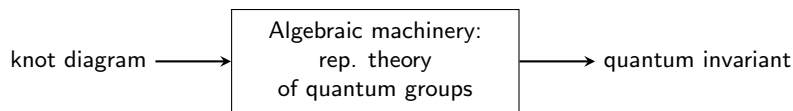


Key in the study of low-dimensional topology!

Motivation

Quantum invariants of knots

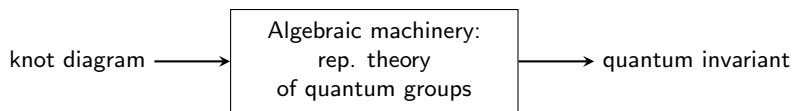
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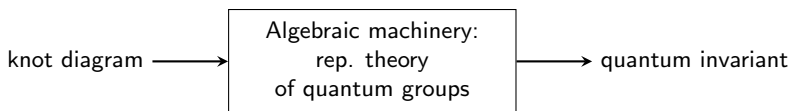
Why should we care about quantum topology?:

- 1 Strong distinguishing power.
- 2 Detects 4-dimensional information, incl. smooth vs. topological sliceness.

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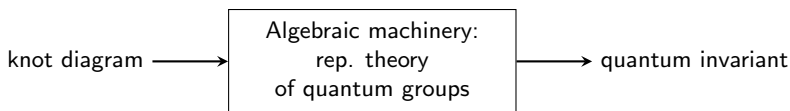
Question: what geometric content is read by quantum knot invariants?

Striking conjectures! $\rightsquigarrow$ volume, incompressible surfaces, genus...

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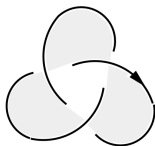
Our work: the $\mathfrak{sl}_2$ semisimple quantum knot invariants read existence of fibration over the knot complement.

Knots and fiberedness

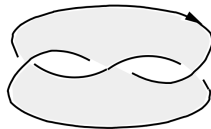
Knots and surfaces

Knot $K \subset S^3 \rightsquigarrow$ connected, cpt, oriented surface S smoothly embedded in S^3 , $\partial S = K$.
We call such a surface a Seifert surface of K .

Minimum genus of such surfaces $\rightsquigarrow g(K)$.



M_0



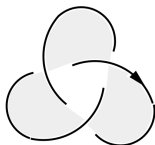
M_1

Typically, a knot has many distinct minimal Seifert surfaces.

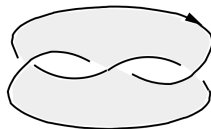
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What can we use Seifert surfaces for? We can construct the *Alexander polynomial* of a knot K .

Example: figure-eight

$$\Delta_{4_1}(x) = x^{-1} - 3 + x$$

Fibered knots

There is a subclass of knots that have a special Seifert surface called fibered knots.

Definition: A knot $K \subset S^3$ is **fibered** if there exists a fibration $S^3 \setminus K \rightarrow S^1$ with fiber F , $\partial F = K$.

We call the closure of F the (unique) fibered surface of K .

Proposition: $g(F) = g(K)$.

Fibered knots

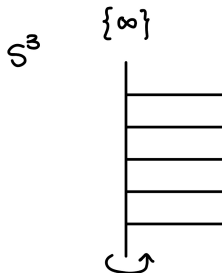
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The Alexander polynomial is the characteristic polynomial of the map induced by the monodromy $h : F \rightarrow F$ on the first homology.

Theorem: Alexander polynomial of a fibered knot is monic and has degree $g(K)$.

$$\Delta_{4_1}(x) = x^{-1} - 3 + x$$

Hopf invariant or enhanced Milnor number

Let $N(K) \simeq D^2 \times K$ be a neighbourhood of K . We define a vector field ξ_K on S^3 :

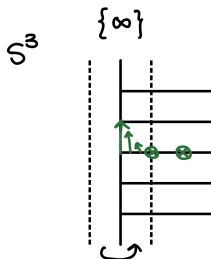
- $S^3 \setminus N(K)$: normal vector to the fibered surface.
- K : oriented tangent vector.
- $N(K)$: interpolation.

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Example: unknot (standard contact structure of S^3).



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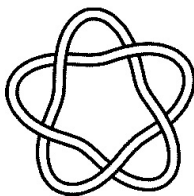
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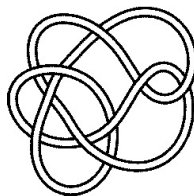
The **enhanced Milnor** number or **Hopf** invariant $\lambda(K)$ is the homotopy class of the plane field ξ_K inside $TS^3 \rightsquigarrow [\xi_K] \in \mathbb{Z}$.

Example: the knots 5_1 and 10_{132} have the same genus and Alexander polynomial, but **different Hopf invariant!**

$$\Delta_{5_1}(x) = \Delta_{10_{132}}(x) = x^{-2} - x^{-1} + 1 - x + x^2$$



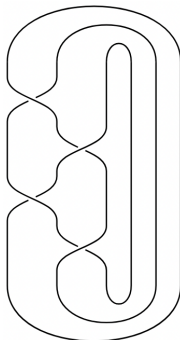
$$\lambda(5_1) = 0$$



$$\lambda(10_{132}) = 2$$

Homogeneous braid knots

An important subclass of fibered knots are *homogeneous braid knots*.



For homogeneous braid knots: simplest Seifert surface is the fibered surface.

This does not hold in general!

Quantum knot invariants associated with $U_q(\mathfrak{sl}_2)$

Review of quantum knot invariants

Quantum invariants associated with the quantum group $U_q(\mathfrak{sl}_2)$:

- Colored Jones polynomials (1984,1990s): $J_n(q) \in \mathbb{Z}[q^{\pm 1}]$.
- ADO polynomials (1992): $N_p(x, w_{2p}) \in \mathbb{Z}[w_{2p}^{\pm 1}][x + x^{-1}]$, where w_{2p} denotes the $2p$ -th root of unity.
- Habiro series (2006): $a_n \in \mathbb{Z}[q^{\pm 1}] \rightsquigarrow H_K(y, q) = \sum_{n \geq 0} a_n(q) \sigma_n(y) \in \Lambda$ where $y = x + x^{-1} - 2$.
- Gukov-Manolescu series (2018): $F_K(x, q) \in \mathbb{Z}((q))[[x]]$. Associated to a physical 3d $\mathcal{N} = 2$ theory.

State sum model

The quantum group $U_q(\mathfrak{sl}_2)$ is a “ q -deformation” of $\mathfrak{sl}_2$. Generated by $\{E, F, K, H\}$ and the set of relations:

$$[H, E] = 2E, \quad [H, F] = 2F, \quad [E, F] = \frac{K - K^{-1}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}}, \quad K^{\pm \frac{1}{2}} = q^{\pm \frac{H}{4}}$$

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and braiding

$$R = q^{\frac{H \otimes H}{4}} \sum_{k \geq 0} \frac{(q^{\frac{1}{2}} - q^{-\frac{1}{2}})^k}{[k]_q} q^{k(k-1)/4} E^k \otimes F^k \in U_q(\mathfrak{sl}_2) \hat{\otimes} U_q(\mathfrak{sl}_2)$$

where $[k]_q := \frac{q^{\frac{k}{2}} - q^{-\frac{k}{2}}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}}.$

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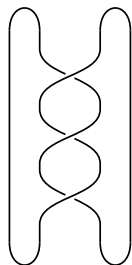
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For generic q : $\{\text{simple f.d. representation } V_n\} \leftrightarrow \{\text{dim. of the vector space } (n+1)\}.$

R acting on a **finite-dim.** representation of $U_q(\mathfrak{sl}_2)$ is a $(n+1) \times (n+1)$ matrix and induces a representation of the braid group $\rightsquigarrow$ we can compute **knot invariants**.

Example of the state sum: trefoil

Example: trefoil. The semisimple invariants (colored Jones) are computed via:


$$\xrightarrow{J_n} \text{tr}_{V_n \otimes V_n}^q \begin{pmatrix} V_n \otimes V_n \\ \uparrow R \\ V_n \otimes V_n \\ \uparrow R \\ V_n \otimes V_n \\ \uparrow R \\ V_n \otimes V_n \end{pmatrix} = \begin{cases} q + q^3 - q^4 & n = 1 \\ q^2 + q^5 - q^7 + q^8 - q^9 - q^{10} + q^{11} & n = 2 \\ \dots & \end{cases}$$

Gukov-Manolescu series

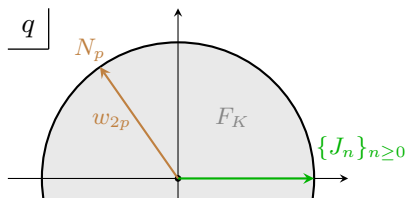
The Gukov-Manolescu series F_K is an **integral power series** in two variables, x and q .

Conjecturally, it recovers the same information as the colored Jones polynomials at $q = 1$:

$$F_K(x, 1+h) = \frac{1}{\Delta_K(x)} + \sum_{i=1}^{\infty} \frac{P_K^{(j)}(x)}{\Delta_K(x)^{2j+1}} h^j.$$

Constantino-Gukov-Putrov also related the F_K invariant to the quantum invariants at root of unity, by conjecturing:

$$F_K(x, w_{2p}) = \frac{N_p(xw_{2p}^{-1})}{\Delta_K(x^r)}.$$



Initial construction of the GM series

Let $V_{q^{-1}x}^h$ be the infinite-dimensional representation with highest weight $q^{-1}x$. For $j \geq 0$:

$$E \cdot v_j = x^{\frac{1}{2}} q^{-j} [j]_q v_{j-1}, \quad F \cdot v_j = x^{-\frac{1}{2}} q^{\frac{j}{2}} [\log_q x - j]_q v_{j+1}, \quad K \cdot v_j = x q^{-j-1} v_j$$

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The R -matrix acting on $V_{q^{-1}x}^h \otimes V_{q^{-1}x}^h$ is:

$$R^{\pm 1} \cdot (v_i \otimes v_j) = \sum_{i+j=i'+j'} (R^{\pm 1})_{i,j}^{i',j'} (v_{i'} \otimes v_{j'})$$

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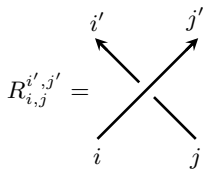
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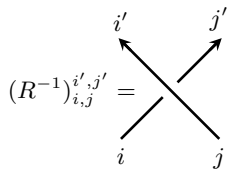
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In pictures:



Positive crossing



Negative crossing

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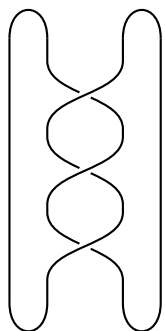
where

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$$(R^{-1})_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{-ii'} (qx)^{-\frac{i+i'+1}{2}} \frac{(q^{-j}; q)_{j-i'}}{(q^{-1}; q^{-1})_{j-i'}} (q^{-i-1}x^{-1}; q^{-1})_{j-i'},$$

$$(a; q)_n := (1-a)(1-aq) \cdots (1-aq^{n-1}).$$

Example: GM state sum of the trefoil knot



$$\xrightarrow{F_K} \operatorname{tr}_{V \otimes V}^q \begin{pmatrix} V \otimes V \\ \uparrow R \\ V \otimes V \\ \uparrow R \\ V \otimes V \\ \uparrow R \\ V \otimes V \end{pmatrix} =$$

$$= qx - q^2 x^3 - q^3 x^4 + q^6 x^6 + q^8 x^7 - q^{13} x^9 - q^{16} x^{10} + q^{23} x^{12} + q^{27} x^{13} - q^{36} x^{15} + \dots$$

Extension to fibered knots

The state sum is an element of $\mathbb{Z}[q^{\pm 1}][[x]]$ whenever K is a positive braid knot.

Main issue: We need the resulting state sum to have only **positive powers** of x .

$$R_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{jj'} (qx)^{\frac{j+j'+1}{2}} \frac{(q^i; q^{-1})_{i-j'}}{(q; q)_{i-j'}} (q^{j+1}x; q)_{i-j'}$$
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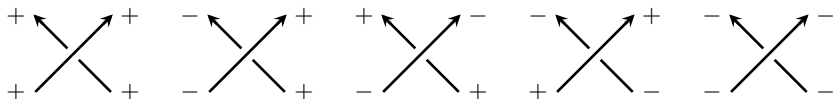
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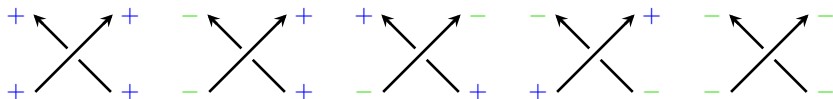
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New R -matrix formulas

extended R -matrix	condition
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$R_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{jj'} (qx)^{\frac{j+j'+1}{2}} \frac{(q^i; q^{-1})_{j'}}{(q; q)_{j'}} \frac{1}{(q^jx; q^{-1})_{j'-i}}$	$j' \geq 0 > i$
$(R^{-1})_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{-ii'} (qx)^{-\frac{i+i'+1}{2}} \frac{(q^{-j}; q)_{j-i'}}{(q^{-1}; q^{-1})_{j-i'}} (q^{-i-1}x^{-1}; q^{-1})_{j-i'}$	$j \geq i' \geq 0$ $0 > j \geq i'$
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The class of nice knots

Definition (Park): A knot K is *nice* if there exists a braid representative of K admitting a $\{\pm 1\}$ -coloring of its segments such that the resulting state sum gives a power series in q and x .

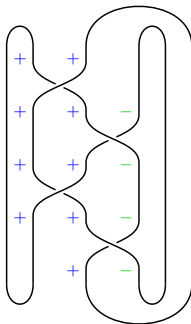
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Example (figure-eight knot):



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Theorem (Park): Let K be a nice knot. Then, its GM series F_K recovers the colored Jones polynomials perturbative expansion at the limit $q = 1$.

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In general: finding the braid representative and sign diagram **is very hard** in practice.

Most braids do not have a sign diagram.

Our work: the GM series and fiberedness

What do these series have in common?

(Besides coming from fibered knots)

$$\begin{aligned}F_{4_1}(x, q) &= x + 2x^2 + (q^{-1} + 3 + q)x^3 \\&\quad + (2q^{-2} + 2q^{-1} + 5 + 2q + 2q^2)x^4 \\&\quad + (q^{-4} + 3q^{-3} + 4q^{-2} + 5q^{-1} + 8 + 5q + 4q^2 + 3q^3 + q^4)x^5 + \dots\end{aligned}$$

$$\begin{aligned}F_{8_{20}}(x, q) &= x^2 + x^3 + (1 - q - q^3)x^4 \\&\quad + (-q - q^3 - q^4)x^5 + (-q^{-1} - q - q^3 - q^4 + q^6 + q^9)x^6 \\&\quad + (-q^{-2} - q^{-1} - q^3 - q^4 + q^5 + q^6 + q^7 + q^9 + q^{10} + q^{11})x^7 + \dots\end{aligned}$$

$$\begin{aligned}F_{10_{153}}(x, q) &= q^{-1}x^3 + (-q^{-3} + q^{-1})x^4 + (q^{-6} - q^{-3} + q^{-1})x^5 \\&\quad + (-q^{-10} + q^{-6} - 2q^{-3})x^6 \\&\quad + (q^{-15} - q^{-10} + q^{-7} + q^{-6} + q^{-5} - 2q^{-3} - 1)x^7 + \dots\end{aligned}$$

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Fiberedness and Quantum Knot Invariants

Let K be a fibered knot. Denote by $g(K)$ its genus and by $\lambda(K)$ the Hopf invariant associated to the fiber surface of K .

Theorem (Orland–Saunders–A'Court–S.–Svoboda, In preparation)

Let K be a nice knot. Then, its Gukov-Manolescu series $F_K \in \mathbb{Z}[q^{\pm 1}][[x]]$ satisfies

- 1 The leading term in x is of the form $q^{\ell(K)}$ for some $\ell(K)$.
- 2 If K is a homogeneous braid knot, $F_K(x, q) = q^{g(K) - \lambda(K)} x^{g(K)} + (\text{h.o.t. in } x)$.

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Proof: involved using Farkas' lemma and mathematical optimization.

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Proposition (Orland–Saunders–A'Court–S.–Svoboda, In preparation)

The above is also true for all fibered knots up to 12-crossings. (~ 1000 knots!)

Checking all these knots was **computationally intensive**.

We naturally conjecture it holds for all fibered knots.

From sign diagrams to the Hopf invariant

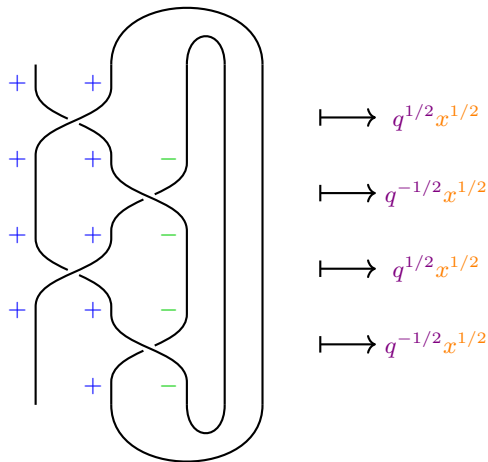
Sign diagrams seem to intrinsically contain **information about the fibered surface** of the knot!

Given a braid and a sign diagram, assign to each of its crossings:

	sign patterns	q -contr.	x -contr.
Positive crossing	$++$, $-+$	$+1/2$	$+1/2$
	$+-$, $-+$	0	0
	$-+$, $+-$		
	$--$	$+1/2$	$-1/2$
Negative crossing	$--$, $+-$	$-1/2$	$+1/2$
	$+-$, $-+$	0	0
	$-+$, $+-$		
	$++$	$-1/2$	$-1/2$

The sum of the q -contributions and x -contributions (with some extra braid-dependent factors) will give $g(K) - \lambda(K)$ and $g(K)$ respectively.

Example: figure-eight



$$q\text{-deg} = \underbrace{0}_{\text{shift}} + 0 = 0 = g(4_1) - \lambda(4_1), \quad x\text{-deg} = \underbrace{-1}_{\text{shift}} + 2 = 1 = g(4_1)$$

An unexpected corollary!

Fiberedness and the colored Jones polynomial

First explicit expression computing the Hopf invariant in terms of colored Jones polynomials!

Corollary (Orland–Saunders–A'Court–S.–Svoboda, In preparation)

$$\lambda(K) = g(K) - \left[\Delta_K^3(y) \cdot \sum_{j=0}^{\infty} \frac{da_j}{dq} \Big|_{q=1} y^j \right]_{y^{2g(K)-1}},$$

where $\{a_j^K(q)\}_{j \geq 0}$ are the Habiro coefficients of the knot K and $\Delta_K(y)$ its Alexander polynomial, whenever K is a homogeneous braid knot.

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Left-hand side: $\lambda(8_{10}) = 4$

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$$N_p^K = x^{-2g(K)(p-1)} q^{2(g(K)-\lambda(K))} + (\text{higher order in } x).$$

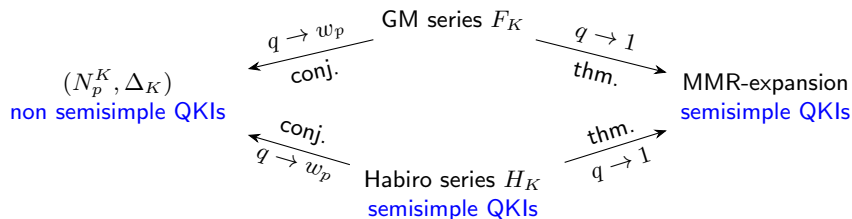
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For K a *nice* knot:



Connection to homological invariants

Connection to knot Floer homology

Knot Floer homology:

$$\widehat{HFK}(K) = \bigoplus_{m,n \in \mathbb{Z}} \widehat{HFK}_m(K, n).$$

Theorem: Δ_K is the graded Euler characteristic of $\widehat{HFK}$.

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$$\widehat{HFK}(10_{132}) = t^{-2}x^{-2} + (2t^{-1} + 1)x^{-1} + (2 + t) + (2t + t^2)x + t^2x^2$$

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Theorem: Let K be a fibered knot. Then,

$$\widehat{HFK}_m(K, g(K)) = \begin{cases} \mathbb{Z} & m = \lambda(K) \\ 0 & \text{else} \end{cases}$$

Recall: $F_K(x, q) = q^{g(K) - \lambda(K)} x^{g(K)} + (\text{higher order terms in } x)$

Beyond fibered knots

There is a candidate procedure to construct the GM series of **non-fibered strongly quasinegative** knots.

Example: $F_{9_{49}}(x, q) = \sum_{n=0}^{\infty} f_n(q)x^n$

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For such knots, $F_K(x, q) = q^{-g(K)}x^{g(K)} + (\text{higher order terms in } q, x).$

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Relation to knot Floer?

For these knots, we conjecture (checked for $\sim 22,000$ knots):

$$\mu(K) := \min \{m \mid \widehat{HFK}_m(K, g(K)) \neq 0\} = 0$$

Then, $F_K(x, q) = q^{-(g(K)-\mu)}x^{g(K)} + (\text{higher order terms in } q, x)$

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Thank you!

Extra slides

Hopf invariant or enhanced Milnor number

Alt. formal definition: Let ξ_1, ξ_2 be two no-where zero vector fields in general position in S^3 . Let

$$\Delta(\xi_1, \xi_2) := \{p \in S^3 \mid \xi_1(p) = t \xi_2(p) \text{ for some } t > 0\}.$$

Define the enhanced Milnor invariant or *Hopf invariant* of ξ_K as

$$\lambda(\xi_K) \equiv \lambda(K) := lk(\Delta(\xi_K, \xi_{std}), \Delta(\xi_K, -\xi_{std}))$$

where ξ_{std} is the canonical contact structure of S^3 and lk denotes the linking number.

Representation theory at generic q

$$[H, E] = 2E, \quad [H, F] = 2F, \quad [E, F] = \frac{K - K^{-1}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}}, \quad K^{\pm \frac{1}{2}} = q^{\pm \frac{H}{4}}$$

$$R = q^{\frac{H \otimes H}{4}} \sum_{k \geq 0} \frac{(q^{\frac{1}{2}} - q^{-\frac{1}{2}})^k}{[k]_q} q^{k(k-1)/4} E^k \otimes F^k \in U_q(\mathfrak{sl}_2) \hat{\otimes} U_q(\mathfrak{sl}_2)$$

Semisimple invariants

For generic $q \in \mathbb{C}$: for every $n \in \mathbb{N}$ there is a simple $U_q(\mathfrak{sl}_2)$ representation V_n of dimension $n + 1$.

$$E \cdot v_j = q^{\frac{n-2j}{4}} [j]_q v_{j-1}, \quad F \cdot v_j = q^{-\frac{n-2j-2}{4}} [n-j]_q v_{j+1}, \quad K \cdot v_j = q^{\frac{n-2j}{2}} v_j$$

$$j = 0, \dots, n.$$

.

Habiro series

Habiro coefficients $a_n(q) \in \mathbb{Z}[q^{\pm 1}]$, $n \geq 0$. Habiro series:

$$H_K(y, q) = \sum_{n=0}^{\infty} a_n(q) \sigma_n(y, q) \in \Lambda = \varprojlim_n \mathbb{Z}[q^{\pm 1}][y]/(\sigma_n) \quad (6.1)$$

where

$$H_K(q^n + q^{-n} - 2, q) = J_n(q) \quad \forall n \geq 0. \quad (6.2)$$

Remark: $\{J_n\} \leftrightarrow \{a_n\}$.

Theorem (Garoufalidis–Bar-Natan, Rozansky, Habiro): There exist polynomials

$\{P_K^{(j)}(y)\}_{j \geq 0} \subseteq \mathbb{Z}[y]$ such that

$$H_K(y, 1 + h) = \frac{1}{\Delta_K(x)} + \sum_{i=1}^{\infty} \frac{P_K^{(j)}(x)}{\Delta_K(x)^{2j+1}} h^j \quad (6.3)$$

where the equality holds when expanding at $h = 0$ and $y = x + x^{-1} - 2 = 0$.